



CYBERTECH

AiBased



COURSE 1 — Robotics & IoT BASIC

Level: Beginner → Intermediate

Fee: ₹15,000

Installment: Available

Module 1 — Robotics Foundation

- **Robotics** কী ও কীভাবে কাজ করে
- **Robot-এর** বিভিন্ন ধরনের ব্যবহার
- **Robot-এর** প্রধান অংশ
- **Controller, Sensor, Actuator, Motor**
- **Robot Block Diagram**

Module 2 — Basic Electronics

- **Voltage, Current, Resistance, Power**

- Ohm's Law
- AC vs DC
- Series & Parallel Circuit
- Resistor, LED, Capacitor, Diode
- Transistor, Relay, Buzzer, Switch
- Practical: LED, Traffic Light, Alarm

Module 3 — Arduino Programming

- Arduino UNO/Nano
- Arduino IDE
- Digital & Analog Pins
- PWM
- Variables & Data Types
- Operators
- if/else
- Loops
- Functions
- Arrays
- Serial Communication

Module 4 — Sensors

- IR
- Ultrasonic

- **LDR**
- **Temperature**
- **Humidity**
- **PIR**
- **Gas**
- **Flame**
- **Soil Moisture**
- **Rain Sensor**

Projects:

- **Automatic Street Light**
- **Fire Alarm**
- **Motion Detector**
- **Smart Plant Monitoring**

Module 5 — Motors

- **DC Motor**
- **Servo Motor**
- **Stepper Motor**
- **Gear Motor**
- **Motor Driver**
- **H-Bridge**
- **PWM Speed Control**

- **Direction Control**

Module 6 — Robot Car

- **Robot Chassis**
- **Motor & Motor Driver**
- **Arduino**
- **Battery**
- **IR Sensor**

Projects:

- **Basic Robot Car**
- **Bluetooth Robot**
- **Obstacle Avoiding Robot**
- **Line Following Robot**

Module 7 — Wireless Communication

- **Bluetooth**
- **Wi-Fi**
- **RF**
- **IR Communication**
- **HC-05/HC-06**
- **ESP8266**
- **ESP32**

Module 8 — IoT Foundation

- IoT कै?
- IoT Architecture
- Sensor → Controller → Internet → Cloud
- Smart Home
- Smart Agriculture
- Industrial IoT

Module 9 — ESP32 & ESP8266

- GPIO
- Wi-Fi
- Bluetooth
- ADC
- PWM
- Web Server
- Remote Device Control

Module 10 — Basic IoT Projects

- Wi-Fi LED Control
- Temperature Monitoring
- Smart Plant Monitoring
- Wireless Sensor
- Basic Smart Home

Basic Final Projects

1. **Line Following Robot**
2. **Obstacle Avoiding Robot**
3. **Bluetooth Controlled Robot**
4. **Smart Irrigation**
5. **Basic Smart Home**

Basic Course Outcome:

শিক্ষার্থী **Arduino/ESP32 programming, electronics, sensors, motors, robot building** এবং **basic IoT system** তৈরি করতে পারবে।

COURSE 2 — Advanced Robotics, IoT & AI

Level: Intermediate → Advanced → Professional

Fee: ₹35,000

Installment: Available

Module 1 — Advanced Robotics Programming

- **Algorithm**
- **Flowchart**
- **Logic Building**
- **State Machine**
- **Feedback System**
- **Sensor Fusion**
- **PID Control**

- Real-time Control

Module 2 — Advanced Autonomous Robot

- Advanced Line Following
- PID Line Following
- Junction Detection
- Multiple Sensor Integration
- Autonomous Navigation
- Path Selection
- Obstacle Detection
- Decision Making

Module 3 — Advanced IoT

- HTTP / HTTPS
- MQTT
- WebSocket
- REST API
- Publisher / Subscriber
- MQTT Broker
- Topics & Messages

Module 4 — IoT Cloud

- Cloud Computing
- IoT Cloud

- **Real-time Monitoring**
- **Data Storage**
- **Blynk**
- **ThingSpeak**
- **Firebase**
- **MQTT Dashboard**

Project:

 **Online IoT Monitoring System**

Module 5 — Mobile + IoT

- **Android IoT Concept**
- **HTTP Request**
- **JSON**
- **API Communication**
- **Real-time Data**
- **Mobile Device Control**

Project:

 **Mobile Controlled Smart Home**

Module 6 — IoT Security

- **Authentication**
- **Authorization**
- **Password Security**
- **Secure Wi-Fi**

- **HTTPS**
- **Secure API**
- **MQTT Security**
- **Device & Firmware Security**
- **Basic IoT Security Testing**

Module 7 — Raspberry Pi

- **Raspberry Pi**
- **Linux Basics**
- **GPIO**
- **Python**
- **SSH**
- **Camera**
- **Sensors**
- **Web Server**

Projects:

- **Motion Detection**
- **Camera Streaming**
- **Raspberry Pi Web Server**

Module 8 — Python for Robotics & IoT

- **Python Fundamentals**
- **Variables**

- **Conditions**
- **Loops**
- **Functions**
- **Lists & Dictionaries**
- **Modules**
- **File Handling**
- **GPIO**
- **Sensor Control**
- **Motor Control**
- **Serial Communication**
- **API Communication**

Module 9 — Computer Vision

- **Image & Pixel**
- **Image Processing**
- **Camera**
- **OpenCV**
- **Motion Detection**
- **Object Detection**
- **Face Detection**

Projects:

- **Smart Security Camera**

- **Face Detection System**
- **Motion Detection Camera**

Module 10 — AI + Robotics

- **AI Fundamentals**
- **Machine Learning Concept**
- **AI vs Automation**
- **Computer Vision**
- **Speech Recognition**
- **AI Decision Making**
- **AI Object Detection**
- **Voice Controlled Robot**
- **Intelligent Automation**

Module 11 — Robotic Arm

- **Servo Motors**
- **Robotic Arm Structure**
- **Controller**
- **Gripper**
- **Pick & Place**
- **Object Sorting**
- **Sensor-based Pick & Place**
- **Vision-based Object Detection**

Module 12 — Smart Agriculture

- **Soil Moisture**
- **Temperature**
- **Humidity**
- **Rain Sensor**
- **Light Sensor**
- **Automatic Irrigation**
- **Automatic Water Pump**
- **Remote Farm Monitoring**
- **Mobile Monitoring**

Module 13 — ROS & Autonomous Robotics

- **ROS Introduction**
- **ROS Architecture**
- **Nodes**
- **Topics**
- **Messages**
- **Services**
- **Packages**
- **Launch Files**
- **Basic Simulation**
- **SLAM Concept**
- **LiDAR**

- IMU
- GPS
- Localization
- Mapping
- Path Planning

Module 14 — Industrial Automation

- Industrial Automation
- PLC Concept
- Conveyor System
- Robotic Arm
- Pick & Place
- Automatic Sorting

Advanced Final Projects

শিক্ষার্থী নিজের পছন্দ অনুযায়ী একটি **Major Project** তৈরি করবে:

-  AI Object Detection Robot
-  AI Security Camera
-  Robotic Arm
-  Smart Agriculture System
-  Mini Industrial Automation
-  Autonomous Robot
-  Advanced IoT Smart Home

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Basic Course = “Build Your First Robot & IoT Device”

Advanced Course = “Build AI-Powered Autonomous Robots & Real-World IoT Systems”

☎ ***Call / WhatsApp:* *9563366057***

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👉 আগে Demo Class দেখুন

👉 Course Details ও Practical Lab সম্পর্কে জানুন

👉 তারপর নিজের ভবিষ্যতের জন্য সঠিক সিদ্ধান্ত নিন!

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